ETH zürich

D-PHYS, ETH Zürich

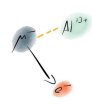
Operating the Mu3e SciFi Detector: Control, Monitoring and Calibration

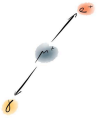
Giacomo Moretti
Supervisor: Marius Köppel
S3P3 Final Event, Bern
September 4th, 2026

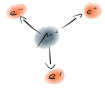
Why search for $\mu \to eee$ – and why it is hard

Why search

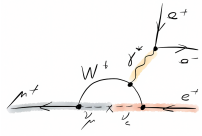
- Charged-lepton flavour violation: **never observed.**
- $\mu^+ \to e^+e^-e^+$ is one of the "golden" channels.
- Yet flavour is violated for neutrinos: $\nu_\mu \to \nu_e$.





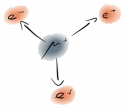






Signal:

$(\mu^+ \to e^+e^-e^+)$ signature



Background:

Internal conversion background
$\mathcal{B}(\mu \to eee\nu\nu) = 3 \times 10^{-5}$



Rejected with excellent **energy** and **momentum** resolution

Accidental background
$(\mu \to e\nu\nu) + (? \to ee) \propto N$



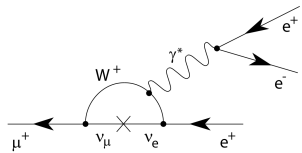
Rejected with excellent **timing** and **vertex** resolution

Why it is hard

- Continuous 10^8 μ/s beam.
- Internal conversion $\mu \to eee\nu\nu$: beat with **energy** + **momentum** resolution.
- Accidental combinations $\propto$ rate: beat with **timing** + **vertex** resolution.

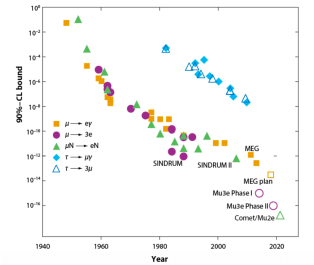
Four orders of magnitude

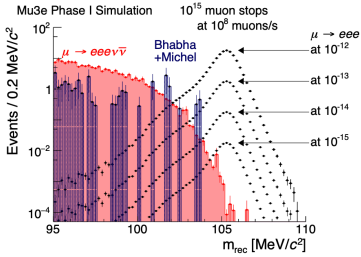
- SM: $\mu \to eee$ only via a ν loop $\Rightarrow$ BR $\sim 10^{-54}$, background-free.
- Any signal = unambiguous new physics.
- SINDRUM $< 10^{-12} \to$ Mu3e 10^{-16}: four orders of magnitude.

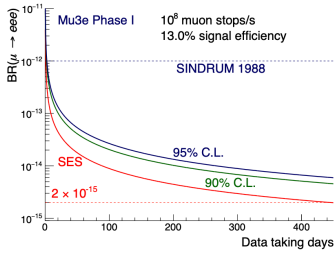


$$\sim \left(\frac{\Delta m_\nu^2}{M_W^2}\right)^2 \Rightarrow BR\left(\mu^\pm \to e^\pm e^+ e^-\right) < 10^{-54}$$

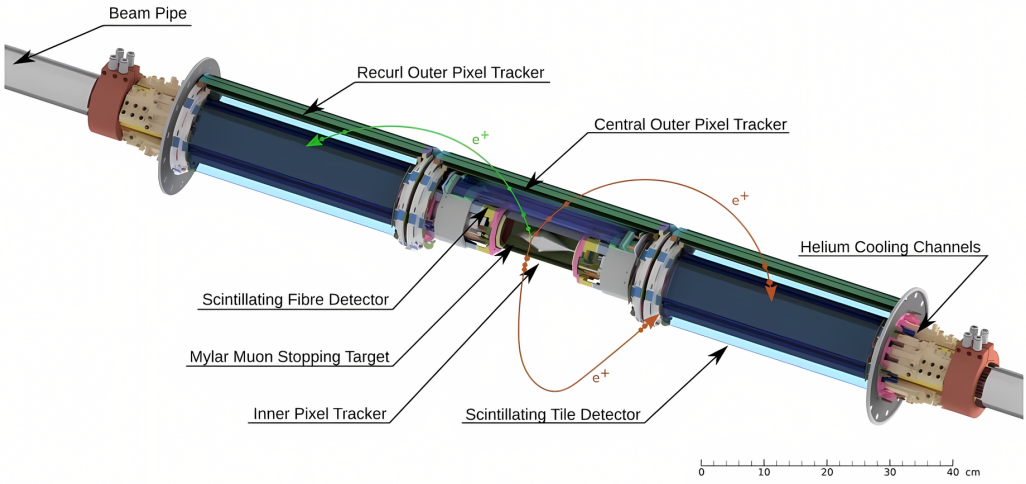
$$\frac{\Gamma\left(\mu^+ \to e^+ e^- e^+\right)}{\Gamma\left(\mu^+ \to e^+ \bar{\nu}_\mu \nu_e\right)} < 10^{-12} \quad (90\%\ \mathrm{CL})$$



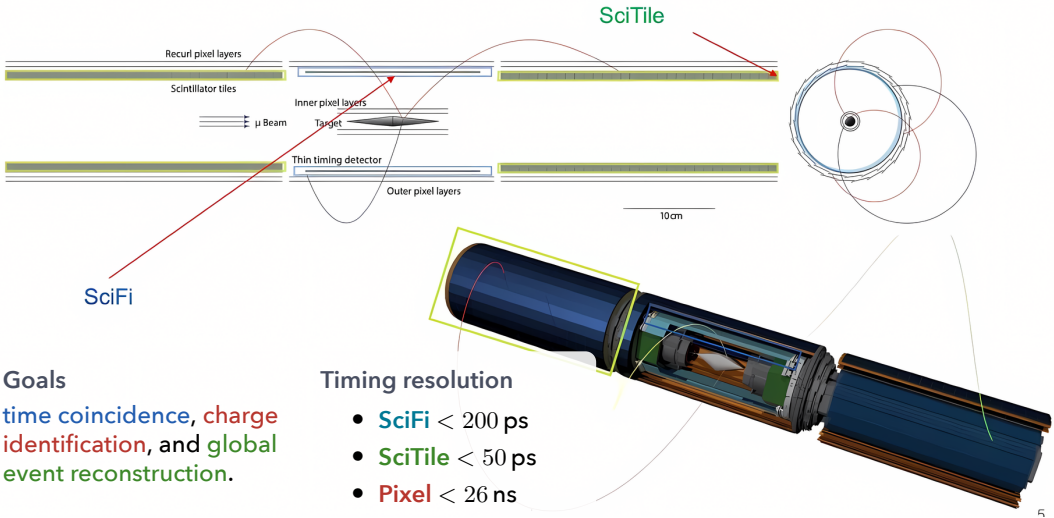




Experimental strategy



Timing layers in Mu3e



Goals

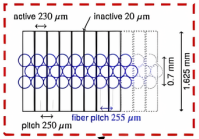
time coincidence, charge identification, and global event reconstruction.

Timing resolution

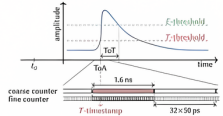
- **SciFi** < 200 ps
- **SciTile** < 50 ps
- **Pixel** < 26 ns

The SciFi detector

Fiber and SiPM array attached



SiPM readout waveform

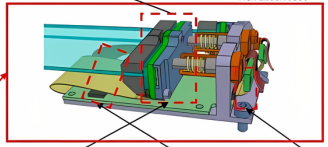


on one end of the fiber ribbon

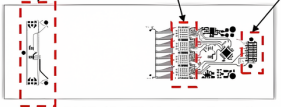
Ref: 2208.09906

Fill factor >60%
μtwsited pair (μTP) cable bundles



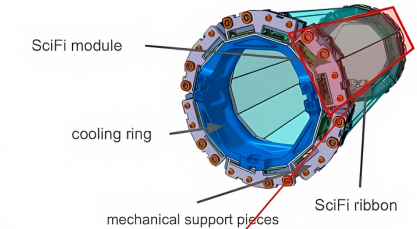


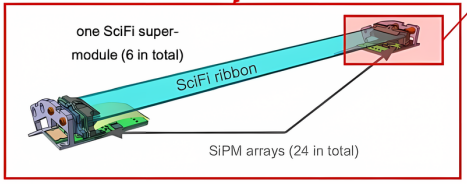
SiPM MuTRiG x4 interposer



SMB (SciFi Module Board)







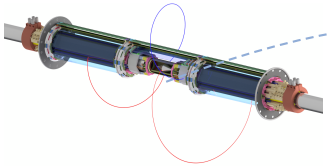
When I started

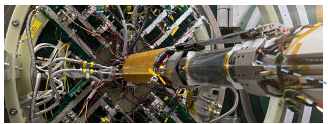
That's the physics. **The rest is my own work**: the SciFi control system, built end to end for the PSI run this October.

- Old page `Scifi-monitor.js`: grew organically, worked, but **hard to trust**, hard to extend.
- Replaced end to end: **Python scripts** driving the detector + a new **custom page** a shifter can rely on.
- What follows: **installation**, **interface**, **configuration**, **calibration**, **saving**.



Where we were → where we are





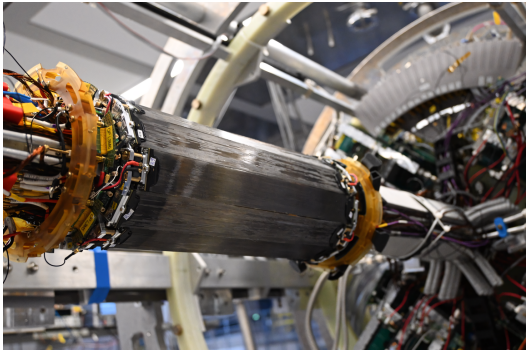


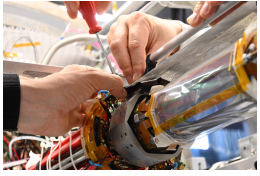
From an **ad-hoc JavaScript page** and a handful of disconnected scripts to a **complete control and monitoring system** – built one floor at a time, this is the path the rest of the talk follows.

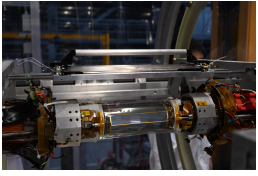
Installation



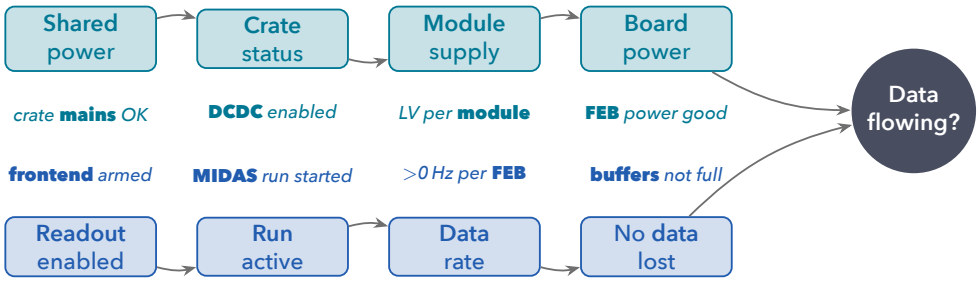
A week of installation at PSI





Two chains, one question



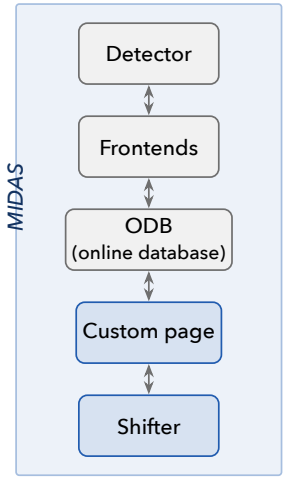
Two chains, checked separately – power can be all green while readout stays idle.
Only when both agree does data actually flow.

User Interface

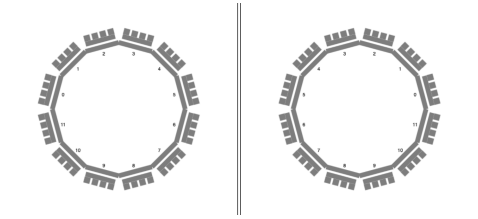


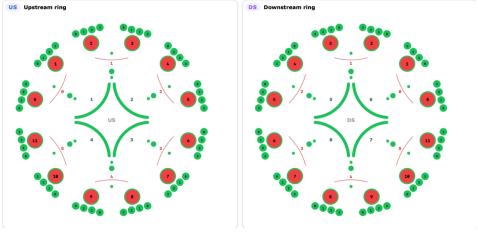
From hardware to shifter

- The chain runs **both ways**: status flows up to the shifter, **changes** flow back down – some only apply at a trigger point (e.g. run start).
- **Nothing is ever hard-blocked**: every action stays clickable; the page warns instead of stopping you.
- **MIDAS** ties it together – frontends, ODB and this page all run inside it.



The page: then and now



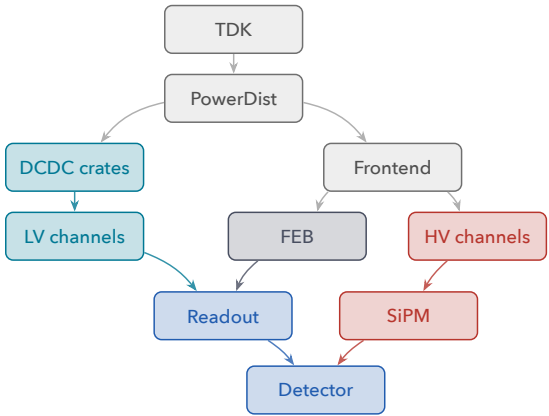


- Old page: a static, unlabeled ring – **no live status**, nothing clickable.
- New page splits in two: **Geometry** read-only diagnostic; **Power** / **Readout** / **Calibration** the only editable surface.
- Each value goes to **ODB** – a frontend pushes it onto the hardware.
- Safety logic stays in **Python**, never duplicated in JavaScript.

Power



From detector to Power and Readout

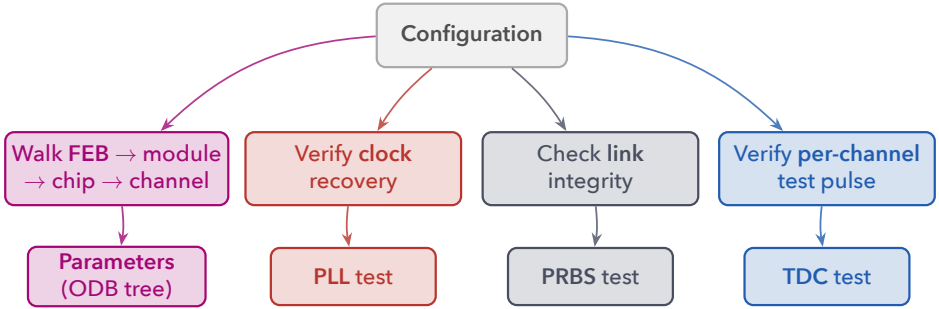


- TDK and PowerDist – the same first two devices for all three chains.
- LV – through its own DCDC crates – feeds the Readout.
- HV and FEB both wait for their Frontend.
- HV feeds the SiPM; FEB feeds the Readout too.
- Readout and SiPM both lead back to the detector.

Configuration

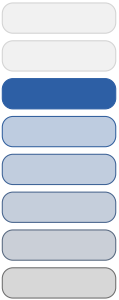


Configuration: one hub, four jobs

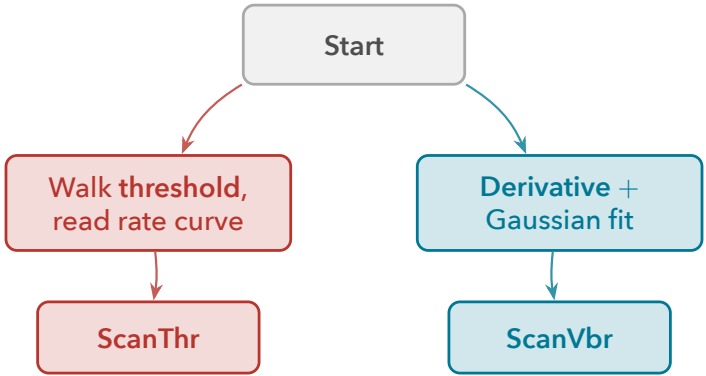


Configuration has grown beyond parameter editing into the detector's general hardware test hub – the old **Readout** tab folds in here entirely, and every parameter still writes to a dedicated **ODB** tree, read directly by the Python scripts.

Calibration



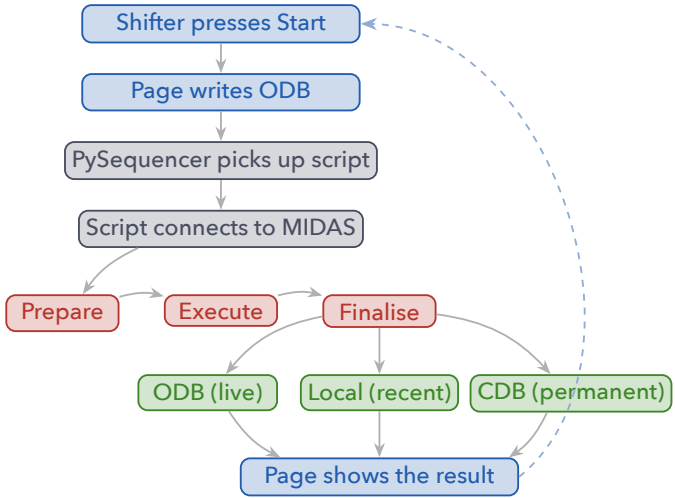
Two calibrations, one starting point



Both share the same `CalibrationBase` lifecycle – **prepare**, **execute**, **finalise**.

How a calibration actually runs

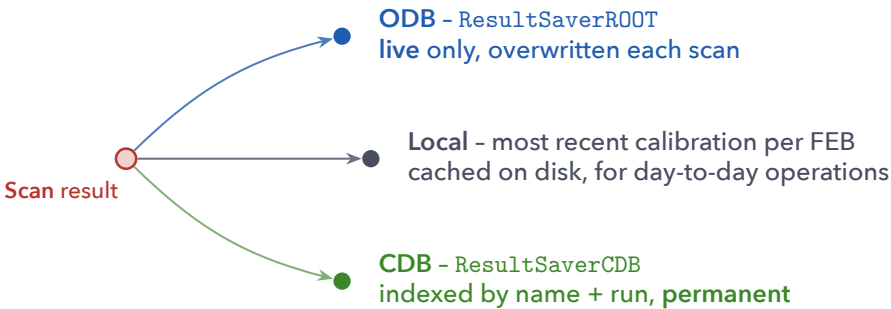
- **Start** – the page writes ODB and hands off to PySequencer.
- **Script** – PySequencer launches it, it connects to MIDAS.
- **Lifecycle** – prepare, execute, finalise.
- **Save** – result written to ODB, local, and CDB.
- **Result** – the page shows it back to the shifter.



Saving



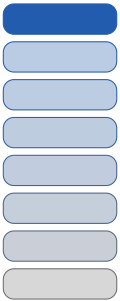
Saving: ODB, local, and CDB



All three destinations are genuinely wired into the calibration scripts now, which was not true before.

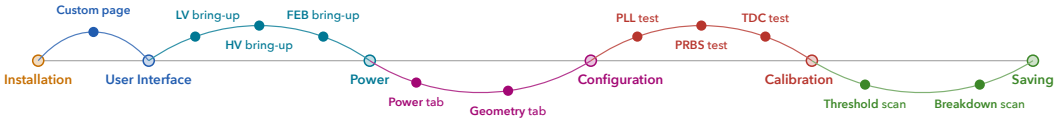
- **ODB** – *"What's the calibration right now?"*
- **Local** – *"What did we last calibrate, for operations?"*
- **CDB** – *"What was it on a given run?"*

Big Picture



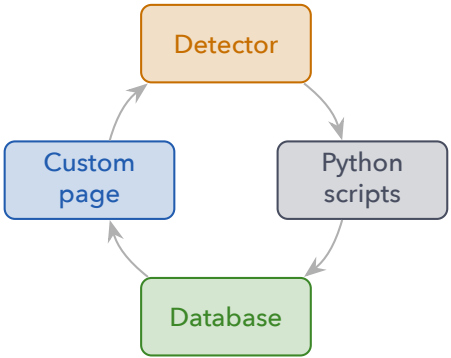
The work, as a branch diagram

It helps to think about this effort the way you'd think about a software repository.



A single **main line** from **Installation** to **Saving**, with every concrete piece built as a **branch**: three **bring-up** sequences, a family of **hardware tests**, two **calibration scans**. Each is built and tested on its own, then merged once solid – converging onto one coherent system.

How it all fits together



- **Detector** – matters once a script can drive it.
- **Python scripts** – matter once the result is saved.
- **Database** – matters once the page shows it back.
- **Custom page** – matters once a shifter can act on it.

This is a control and monitoring system: no single piece works alone. Every step in this loop matters equally.

ETH zürich

Giacomo Moretti

Mu3e Experiment
Scifi Detector Group
ETH Zürich
PSI, Villigen



References

1. Mu3e Collaboration, *Technical design of the Phase I Mu3e experiment*, arXiv:2009.11690 (2021).
2. A. Bravar *et al.*, *Development of the Scintillating Fiber Timing Detector for the Mu3e Experiment*, arXiv:2208.09906 (2022).
3. L. Vigani *et al.*, *The Mu3e detector*, JINST **17** (2022) C05024.
4. M. Köppel (Mu3e Coll.), *Mu3e Integration Run 2021*, PoS(NuFact2021), arXiv:2203.07855 (2022).
5. C. Martín Pérez (Mu3e Coll.), *Search for the LFV decay $\mu^+ \to e^+e^-e^+$ with the Mu3e experiment at PSI*, New Frontiers in Lepton Flavor, Pisa (2023).
6. R. M. Amarinei (Mu3e Coll.), *Scintillating Fibre Detector for the Mu3e Experiment*, NuFact 2024, arXiv:2501.14692 (2025).
7. R. M. Amarinei (Mu3e Coll.), *The Mu3e Experiment: Status and Short-Term Plans*, NuFact 2024, arXiv:2501.14667 (2025).
8. A. Bravar (Mu3e Coll.), *The Mu3e Experiment @ PSI*, Tau 2016, Beijing (2016).
9. Y. Wang (Mu3e Coll.), *The DAQ system of the Mu3e SciFi detector*, SPS/APS Joint Annual Meeting (2023).
10. S. H. Aulich, *Online Anomaly Detection for Mu3e*, ETH Zürich summer-student report (2025).

Backup

Geometry tab: drill down, three clicks deep

SM health dot

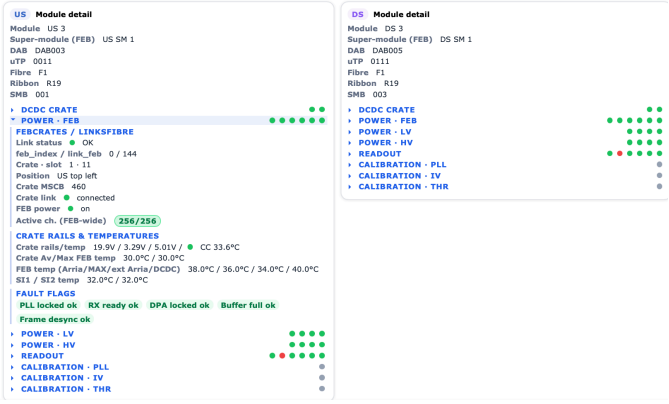
adds HV/Readout/
Calibration status

Module dot

expands the
32-channel grid

ASIC dot

US/DS are drawn as mirror rings with opposite rotation. Each click level is **cumulative**, not a page switch.



What's hardcoded vs. what's live

SAFE	Compile-time constants
SELF-HEALING	Name-pattern filters (`"SMB_"`, `"SciFi CH"`)
REAL RISK	Curated lists: `SciFi_Programs.js` (7), `SciFi_Alarms.js` (10), `HV_BOARDS` (32-cell board↔module map)
LIVE	Everything else

`HV_BOARDS` is the single biggest hardcoded piece on the whole page – flagged, not yet ported into Python/ODB.

Calibration entry points – a deliberate simplification

`thr_run.py` / `vbr_run.py` / `pll_run.py` bypass SciFi's own `CalibrationFactory` (never wired to a real entry point) and instead:

- build `Detector` + `Calibration` **directly**
- open their **own** `MidasClient`, with `try/finally: disconnect()`

Judged safer than pixel's own pattern

Pixel passes PySequencer's `seq` object in as the client – never independently verified. SciFi's explicit connect/disconnect is a stronger guarantee.

From design to a real save

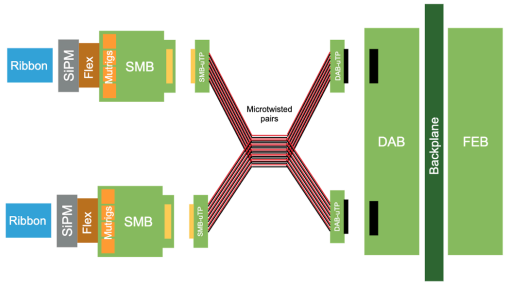
`.root` layout

5 trees – `Meta`, `RawScan`, `Results`, `Conditions`, `ChannelMasks` – the shape was restructured once per-module/per-ASIC parameters were properly pinned down.

`ResultSaverROOT` (per-tree upsert by key), `ResultSaverCDB` (opt-in POST to `mu3edb0/detcal`), and a local cache of the most recent per-FEB calibration are implemented and wired, reconciled with a colleague's parallel branch.

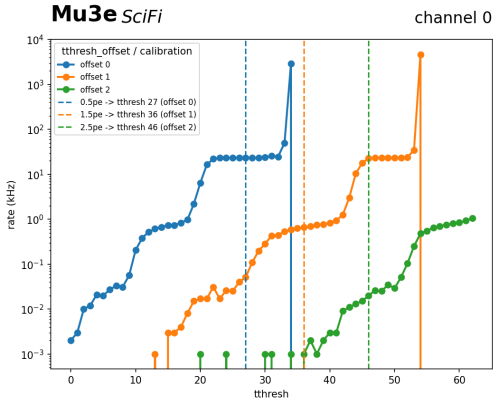
The SciFi detector, briefly

- **2 rings**, US / DS – 12 modules each; **24 SMB** (12+12) merge into 12 super-modules via a passive DAB.
- **12 FEB** (1:1 with LV); **96 ASIC** (32 ch. each) = **3072 channels**; **24 HV channels** (1:1 with SMB).
- Each module: {side, ring position, readout end} – the same key used across **ODB**, **geometry**, and **calibration**.
- SMB and DAB connect via microtwisted-pair links, carrying the raw **MuTRiG3** data downstream.



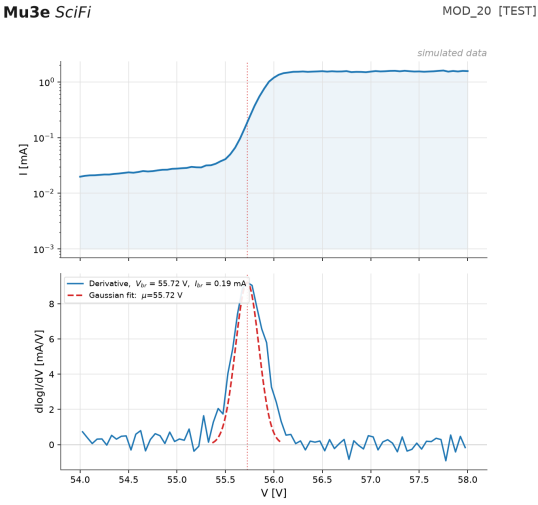
Calibration: threshold scan

- **ScanThr** finds the right discrimination point on every channel.
- Walks the **threshold** up per channel, reading the rate curve to find the crossing points at 0.5, 1.5 and 2.5 p.e.
- Runs independently per **FEB** – any subset of the detector can be (re-)calibrated on its own, and the result is what actually gets loaded as the operating threshold.



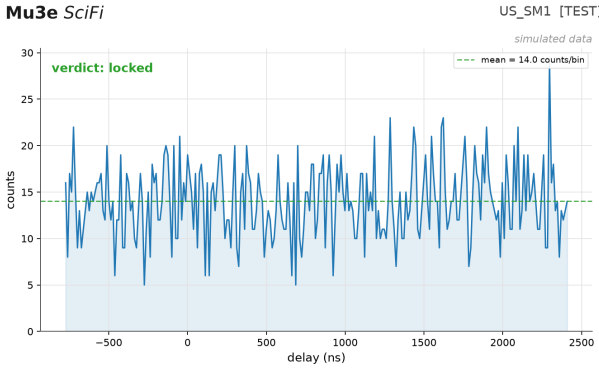
Calibration: IV scan

- **ScanVbr** locates each SiPM's breakdown voltage.
- Takes the **derivative** of the IV curve and fits a **Gaussian** to its peak – the fit mean μ gives V_{br}.
- Runs independently per **FEB** – the result directly sets each SiPM's HV operating point (overvoltage), a real physics constant, not just a diagnostic.



Configuration: PLL test

- The **PLL test** verifies every SMB recovers the clock from its data link.
- It histograms the recovered-clock **delay**: a genuinely **locked** PLL gives a **flat** distribution, an unlocked one a sharp peak.
- Pass/fail comes from the histogram **flatness** (mean vs. spread) – one verdict per SMB.



What this leaves behind

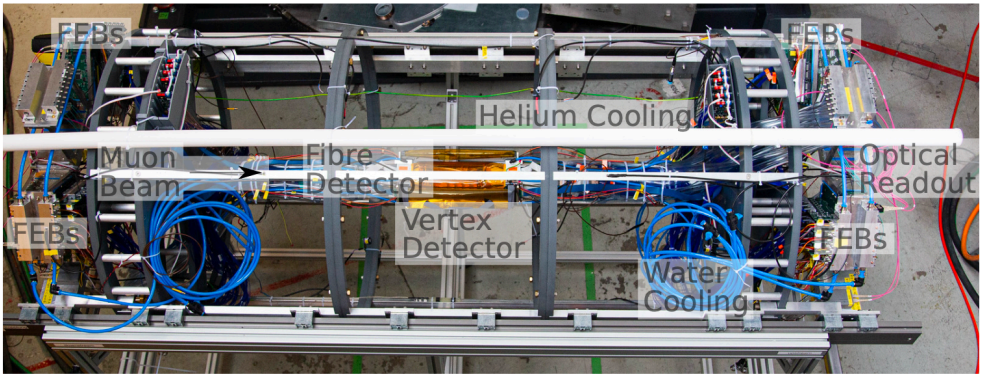
- **Result saving** is fully implemented and wired – the one piece that had to be ready for the data taking, and it is.
- The **shared code library** now properly covers SciFi, closing the gap with pixel: a real, reusable control and monitoring foundation, not a one-off.
- The **PLL verdict logic** was corrected against real data – a good example of the kind of mistake that only shows up once you actually run the thing.
- **Configuration** grew into the detector's general hardware test hub, absorbing the old Readout tab entirely.

Installed and working on `mu3ebe`: LV, HV, FEBs, Readout

What's still open

Everything is installed, wired, and ready. The one thing left is a full end-to-end workflow test, including a real calibration run, on `mu3ebe`.

The detector, today



A SciFi ribbon

